FINANCIAL FRAUD DECTECTION



PRESENTED BY: JARREN MEREDITH

Overview

Secure Guard Financial Solutions — real-time fraud detection

Who We Are

A provider of fraud detection analysis built for the financial industry.

Cutting-edge solutions that safeguard client assets in real time.

Our Mission

Comprehensive fraud detection that helps institutions keep customer trust.

Minimize financial losses from fraudulent transactions.

How We Help

Transaction data is screened as it flows, not after the fact.

Institutions act on suspicious activity immediately.

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The Problem

Fraud risk in a digital-first payments era

Growing Exposure

Online transactions and digital payments keep rising.

Fraud is now a top concern for institutions worldwide.

Many Attack Forms

Unauthorized credit card transactions.

Unusual spending patterns and fraudulent card usage.

Identity theft targeting cardholders.

What Is At Stake

Direct financial losses for cardholders and institutions.

Erosion of customer trust and confidence.

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Our Approach

From transaction data to immediate action

01 Analyze

Study patterns in transaction data across amount, location and behavior.

Build a baseline of what normal looks like.

02 Detect

Flag anomalies that deviate from expected spending patterns.

Score transactions for fraud risk in real time.

03 Act

Surface suspicious transactions to the institution instantly.

Enable immediate action to mitigate the risk.

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Statistical Summary

Transaction amount and city population — 389,002 records

MEAN AMOUNT
$70.44
median $47.57

STD. DEVIATION
$162.20
range $1 – $27,390

MEAN CITY POP.
88,681
median 2,456

TOTAL AMOUNT
$27.40M
sum of all records

Measure	Transaction amount	City population
Mean	70.44	88,680.84
Standard error	0.26	482.94
Median	47.57	2,456
Mode	1.04	606
Standard deviation	162.20	301,210.10
Skewness	40.27	5.59
Minimum / Maximum	1.00 / 27,390.12	23 / 2,906,700
Confidence level (95%)	0.51	946.55

What the numbers say

- Amounts are highly right-skewed: mean $70 sits well above the $48 median.
- Extreme kurtosis (3,868) signals rare, very large transactions.
- City population spans 23 to 2.9M, so most transactions come from small towns.

Source: Book1.xlsx — Statistical Summary

Statistical Summary

Numerical Variable Analysis

1. amt (Transaction Amount):

The amt column exhibits a **highly right-skewed and leptokurtic (heavy-tailed) distribution**, indicating a large number of small transactions and a few extremely large transactions. This is typical in financial data, where fraudulent transactions often involve higher amounts.

Shapiro-Wilk and Anderson-Darling tests strongly reject the hypothesis of normality, confirming its non-normal distribution.

Outliers are present, particularly on the higher end, which is expected given the skewness and is highly relevant for fraud detection.

2. city_pop, lat, long:

These columns also show **non-normal distributions**, with varying degrees of skewness and kurtosis. For instance, city_popis also right-skewed with significant outliers. This suggests that basic statistical models assuming normality might not be appropriate without prior transformations.

3. Correlation Matrix:

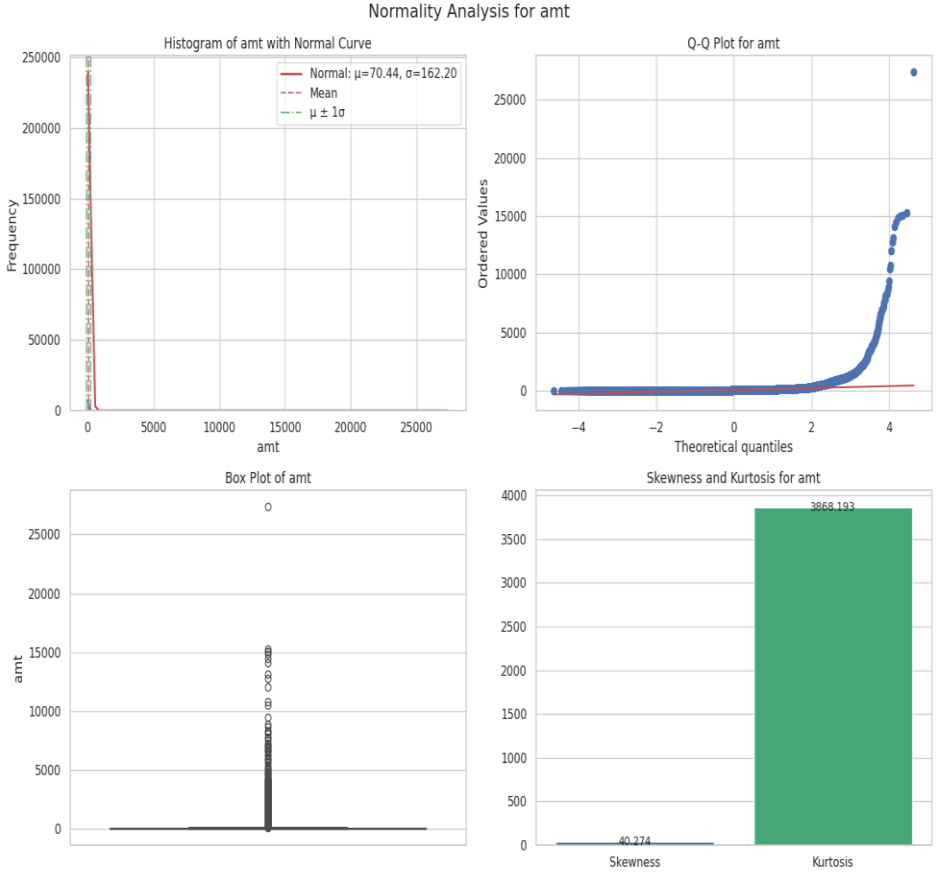
The correlation matrix for amt, city_pop, lat, and long shows **very weak correlations** between these numerical features. This implies that these variables are largely independent of each other in terms of linear relationships.

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Statistical Summary

Mean $70.44 vs. σ $162.20, with the histogram collapsing into a single spike near zero — most transactions are very small.
Q-Q plot curves sharply upward at the right tail, so values deviate strongly from normal.
Box plot shows a dense column of high-end outliers, up to roughly $27,000.
Kurtosis ≈ 3,868 and skewness ≈ 40.3 — extreme heavy-tailed, right-skewed behavior.

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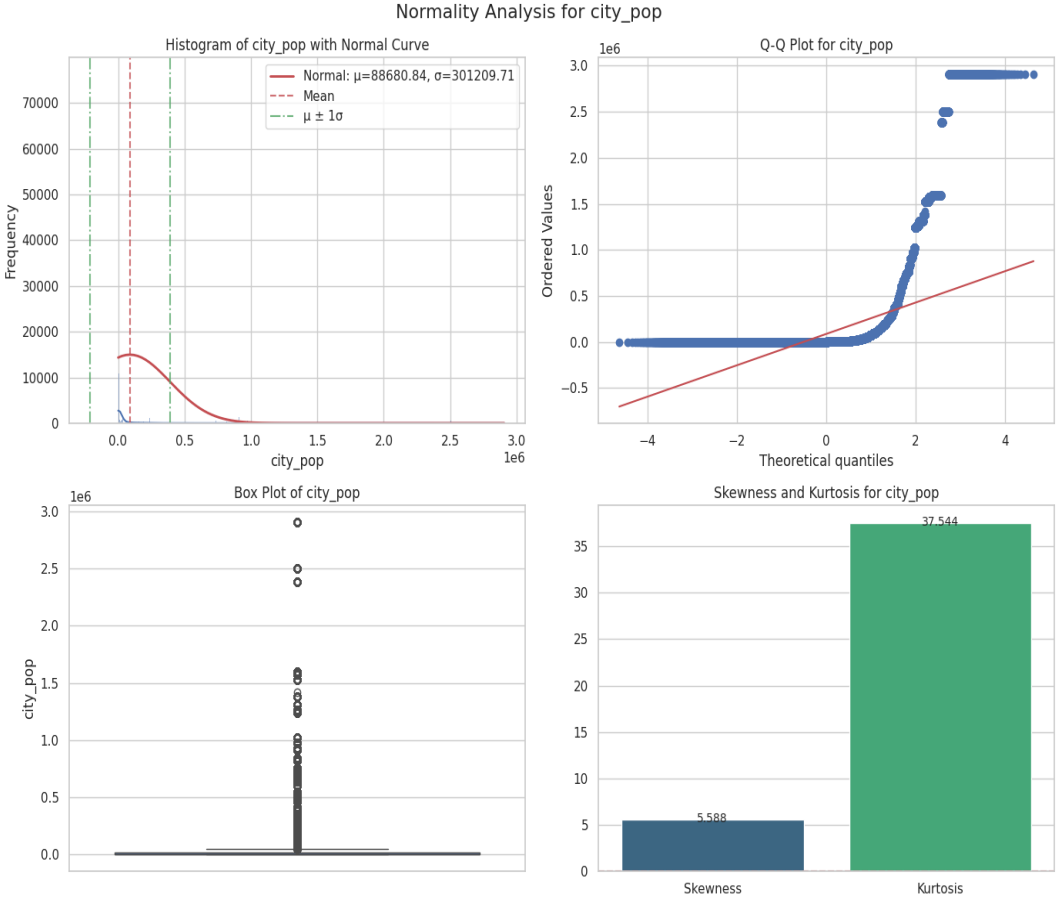


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Statistical Summary

Mean 88,681 vs. σ 301,210 — variation far larger than the average, driven by a few big metros.
Histogram and box plot concentrate near zero with outliers reaching about **2.9M**.
Q-Q plot again departs from the reference line, confirming non-normality.
Skewness ≈ 5.6, kurtosis ≈ 37.5 — skewed, but much milder than transaction amount.

Bottom line: Neither variable is normally distributed, and both are dominated by rare extreme values — which is why transformations or outlier-aware models matter, and why the large-amount tail is exactly where fraud signals tend to live.



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Top 3 States By Transaction Volume

Headline: Texas, New York, and Pennsylvania generated **77,427 transactions—19.9% of all 389,002 transactions**.

Key takeaway: Texas leads New York by **3,448 transactions (13.8%)**. Pennsylvania maintains a strong third place, recording **40.6% more transactions than fourth-ranked California**.



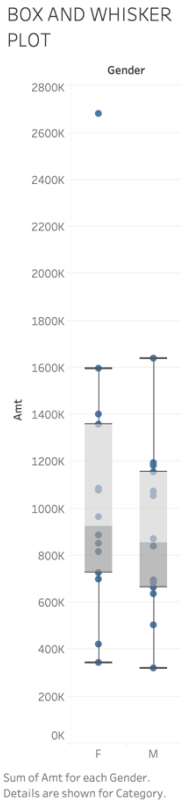
Rank	State	Transactions	Share
1	Texas	28,482	7.3%
2	New York	25,034	6.4%
3	Pennsylvania	23,911	6.1%

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FRAUD BY GENDER AND CATEGORY

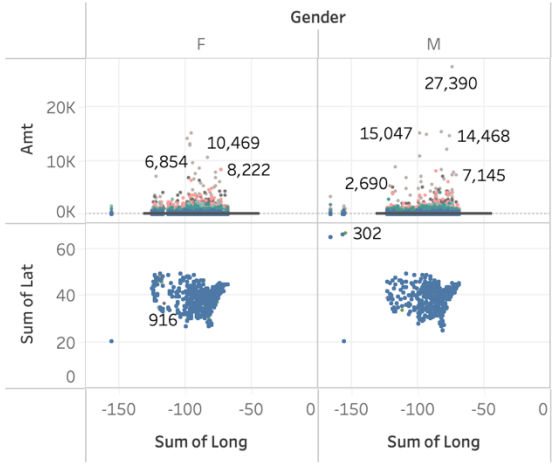
Fraud by Gender & Category

2,252 total fraud cases, split nearly evenly: **1,141 female (50.7%)** and **1,111 male (49.3%)**.
Grocery POS was the largest category with **518 cases (23.0%)**, followed by **Shopping Online** with **478 (21.2%)**.
The top five categories accounted for **76.8%** of all fraud.
Shopping Online had the largest gender gap: **290 male vs. 188 female cases**.
Females led in **9 of 14 categories**, although males recorded more cases in several high-volume categories.
Key takeaway: Fraud volume is balanced by gender overall, but prevention efforts should prioritize **grocery POS, online shopping, miscellaneous online, shopping POS, and gas/transport**.

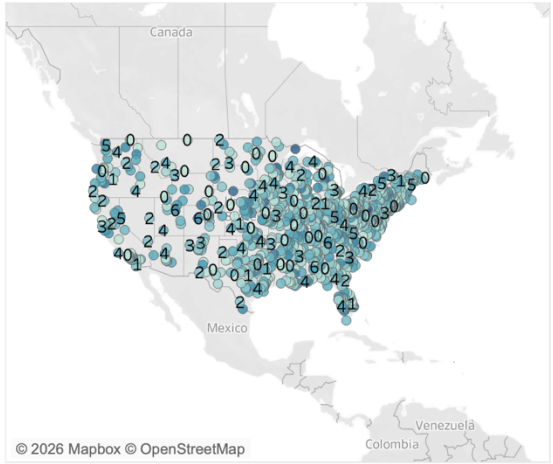


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SCATTER PLOT

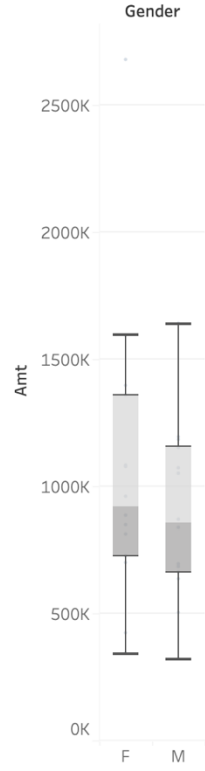


FRAUD MAP

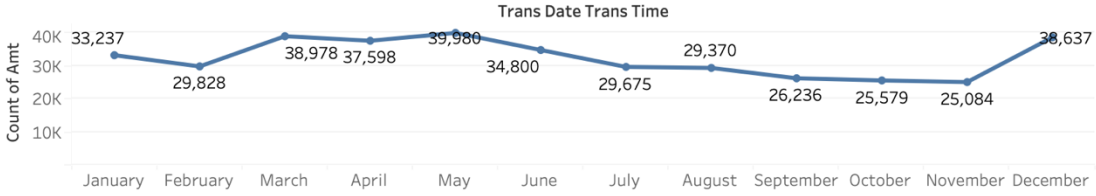


Category
- entertainment
- food_dining
- gas_transport
- grocery_net
- grocery_pos

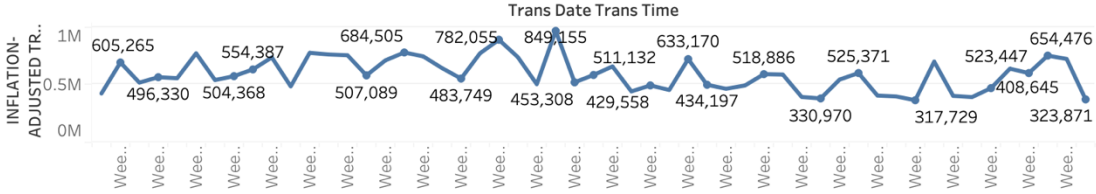
BOX AND WHISKER PLOT



TIME SERIES



ADJUSTED TRANSACTION AMOUNTS (INFLATION)



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System Architecture

A streaming pipeline that scores every transaction in under 300 ms

01 Ingest

Authorization messages stream in from the card network as events.

Enriched with cardholder profile, merchant, device and location data.

02 Score

A rules engine plus a machine learning model return a 0-100 risk score.

Features cover amount, velocity, geography and merchant category.

03 Decide

Approve, step-up authenticate, or decline before the transaction settles.

Every decision is logged for audit and model retraining.

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Detection Signals

What the system watches for on every transaction

01 Unauthorized Use

Activity far from the cardholder's usual geography or device.

Impossible travel: two distant locations within minutes.

02 Unusual Spending

Amounts far beyond the cardholder baseline against a $48 median.

Sudden bursts of activity or brand-new merchant categories.

03 Card Testing

Repeated micro-charges probing whether a stolen card is live.

One card used across many merchants in a short window.

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Response and Performance

Acting on alerts and proving the system works

01 Respond

High risk declines instantly; medium risk triggers a one-time passcode.

Cardholder alerts by app and SMS with a one-tap confirm or deny.

02 Investigate

An analyst queue ranks cases by exposure and confidence.

Confirmed fraud feeds straight back into the model.

03 Measure

Track detection rate, false positive rate and dollars prevented.

Keep customer friction low while losses keep falling.

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Conclusion

Bringing the architecture, signals and response together

01 One Pipeline

Ingest, score and decide runs end to end in under 300 ms, before a transaction settles.

Rules and a machine learning model share the same enriched transaction event.

02 The Right Signals

Unauthorized use, unusual spending and card testing are covered by one scoring layer.

Behavioral baselines make each cardholder their own benchmark, not a global rule.

03 Proven Impact

Graduated responses stop fraud while keeping legitimate customers moving.

Detection rate, false positives and dollars prevented make the value measurable.

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QUESTIONS?

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THANK YOU



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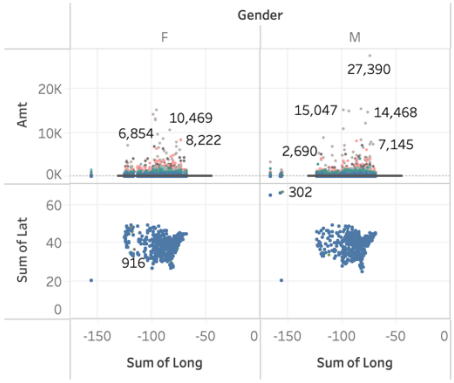
- Datasets: cc_data.csv and location_data.csv

- Analytical programs utilized to perform Data analysis:
  1. Excel
  2. Python
  3. SQL
  4. Tableau

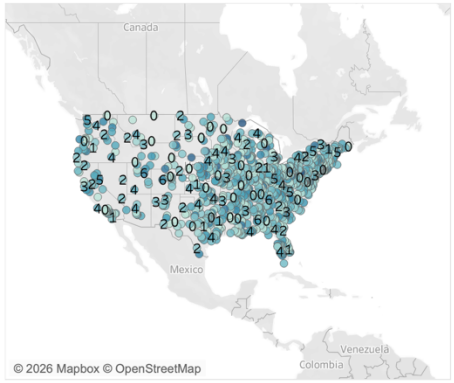
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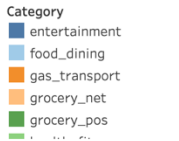
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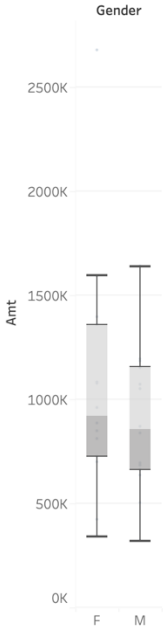


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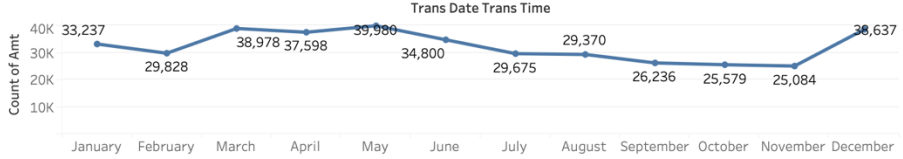




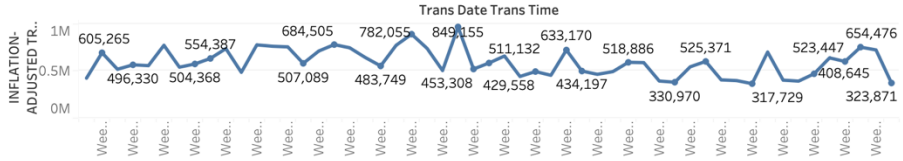
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NAVY
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TRUST BLUE
#5B7A99
STEEL BLUE
#D6D9DD
LIGHT GRAY
#7A7D7F
CHARCOAL GRAY
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